CPL 2025 - FOUR-LEAGUE VALIDATION

# Caribbean Stress Test

Fourth league test - and the first where

the system meets its limits. 191 matches, 4 leagues.

**46.9%**
CPL-TUNED ACCURACY
Tied for 2nd with ELO+Momentum

**50.0%**
LEAGUE WINNER (LOGREG)
First ML win across 4 leagues

**+6.3pp**
LIFT VS BASELINE
Smallest lift of 4 leagues

# Executive Summary

This report completes the four-league validation of the cricket match-prediction engine by applying the same thirteen prediction systems to the Caribbean Premier League 2025 season. The CPL features six franchises (SKNP, ABF, GAW, BT, TKR, SLK) playing 34 fixtures between August 15 and September 22, 2025. Two matches were abandoned without a result (Match 5: ABF vs SLK on August 18; Match 12: SLK vs BT on August 25), leaving 32 playable matches for evaluation.

**This is the first league where the Optimized-Weighted Ensemble did not win outright.** The CPL-tuned variant tied for second place with ELO+Momentum at 46.9% accuracy, while Logistic Regression - an ML model that had underperformed on all three previous leagues - won with 50.0%. The CPL is the smallest, highest-variance league tested, and the result reveals the boundary conditions of the methodology.

| 46.9% | 4 / 4 | 50.0% |
|---|---|---|
| CPL-TUNED ACCURACY | LEAGUES WHERE OPT-WEIGHTED RANKS TOP 3 | WINNER (LOGREG) |

The honest finding: **the Optimized-Weighted architecture remains the best overall system across the four-league sample** (it won IPL, PSL, and BBL, and tied for 2nd on CPL), but the CPL exposes its limits. With only 32 matches and 6 teams, prediction accuracy collapses to near-baseline for every system tested. No system beat 50% by a meaningful margin - the theoretical ceiling for this league appears to be around 50-55%, far below the 63-67% achieved on the larger leagues.

This is itself a useful finding: it tells us where the methodology stops working. Cricket prediction at seasonal scale requires either a longer season (more matches per team) or pooling multiple seasons of data. The CPL's six-team, 32-match format is below the viability threshold for any current statistical approach.

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02 - INPUTS

# Data & Methodology

## 2.1 CPL 2025 Source Data

The full 34-match CPL 2025 fixture list was used: 30 regular-season matches (August 15 to September 15) plus four playoff fixtures (Eliminator 1 on September 17, Qualifier 1 on September 18, Qualifier 2 on September 20, and the Final on September 22). Two league matches were abandoned without a result and excluded, leaving 32 playable matches. The CPL has the smallest sample of the four leagues tested and the fewest teams (six), making it the most stressful test of methodological robustness.

## 2.2 Four-League Sample Sizes

| League | Season | Matches | Teams | Matches/Team | Tournament Winner |
|---|---|---|---|---|---|
| IPL | 2026 | 73 | 10 | ~14 | RCB |
| PSL | 2026 | 43 | 8 | ~10 | PSZ |
| BBL | 2025-26 | 43 | 8 | ~10 | PRS |
| **CPL** | **2025** | **32** | **6** | **~10** | **TKR** |

The CPL has the fewest matches (32) and the fewest teams (6). This is the structural reason for the prediction difficulty: with only ~10 matches per team and 6 teams in the league, every team plays every other team only 2-3 times in the regular season, giving the head-to-head signal minimal time to mature. The ELO rating system has fewer matches to absorb information from, and momentum has fewer games to develop.

## 2.3 Four Variants of the Optimized Ensemble

On CPL we test four Optimized-Weighted variants in parallel: **IPL-tuned**, **PSL-tuned**, **BBL-tuned**, and **CPL-tuned** (weights discovered by grid-searching 1,270 combinations on the CPL backtest horizon). This four-way comparison reveals whether any cross-league weight transfer works on the smallest league, or whether the small-sample regime defeats all weight configurations.

03 - HEADLINE NUMBERS

# CPL Backtest Results

Across 32 walk-forward predictions, no system achieved more than 50% accuracy. Logistic Regression won at 50.0% (12/24 - it only covered 24 matches due to the 4-match warm-up), with the Opt-Weighted (CPL-tuned) and ELO+Momentum tied for second at 46.9%. The IPL-tuned, PSL-tuned, and BBL-tuned variants all fell to 40.6% - exactly at the baseline.

| Rank | System | Type | N | Correct | Acc | Brier | LogLoss |
|---|---|---|---|---|---|---|---|
| 1 | LogReg | ML | 24 | 12 | 50.0% | 0.382 | 1.209 |
| 2 | ELO+Momentum | Rating | 32 | 15 | 46.9% | 0.285 | 0.784 |
| 3 | Opt-Weighted (CPL-tuned) | Blend | 32 | 15 | 46.9% | 0.290 | 0.783 |
| 4 | RandomForest | ML | 24 | 11 | 45.8% | 0.317 | 0.850 |
| 5 | Ensemble-Stacked | ML | 24 | 11 | 45.8% | 0.393 | 1.092 |
| 6 | Opt-Weighted (BBL-tuned) | Blend | 32 | 14 | 43.8% | 0.277 | 0.753 |
| 7 | Bayesian-Shrunk | Blend | 32 | 14 | 43.8% | 0.252 | 0.697 |
| 8 | GradientBoosting | ML | 24 | 10 | 41.7% | 0.544 | 3.790 |
| 9 | ELO-Raw | Rating | 32 | 13 | 40.6% | 0.267 | 0.732 |
| 10 | Opt-Weighted (IPL-tuned) | Blend | 32 | 13 | 40.6% | 0.287 | 0.773 |
| 11 | Opt-Weighted (PSL-tuned) | Blend | 32 | 13 | 40.6% | 0.285 | 0.769 |
| 12 | Baseline-50/50 | Baseline | 32 | 13 | 40.6% | 0.250 | 0.693 |
| 13 | Weighted-Score | Blend | 32 | 12 | 37.5% | 0.279 | 0.755 |
| 14 | Pythagorean | Rating | 32 | 12 | 37.5% | 0.260 | 0.714 |

*Every system is clustered in the 37-50% range - the smallest gap between best and worst across all four leagues. The CPL is genuinely hard to predict.*

## 3.1 Why the CPL Breaks the Pattern

**Reason 1: Sample size.** With 32 matches, the 95% confidence interval on a 50% accuracy is roughly +/- 17 percentage points. The 0.1 percentage point gap between first (LogReg, 50.0%) and second (Opt-Weighted, 46.9%) is well within statistical noise - we cannot confidently say LogReg is actually better on CPL.

**Reason 2: High match variance.** CPL 2025 had an unusually high share of close results and upsets: 7 of 32 matches (22%) were decided by 10 or fewer runs or 2 or fewer wickets. In a high-variance environment, team-level signals (ELO, run rate) carry less information because the outcome is increasingly determined by match-day execution.

**Reason 3: Short, dense schedule.** The CPL runs for 5-6 weeks with 2-3 matches per week per team. Teams rotate squads heavily, resting key players in some matches. The system has no visibility into player availability, so its team-level signals become noisier when squads are shuffled.

**Reason 4: Tropical conditions.** Caribbean pitches and weather are more variable than subcontinental or Australian conditions - rain interruptions (2 abandoned matches this season), dew under lights, and pitch deterioration all increase match-to-match variance.

## 3.2 Visual Comparison

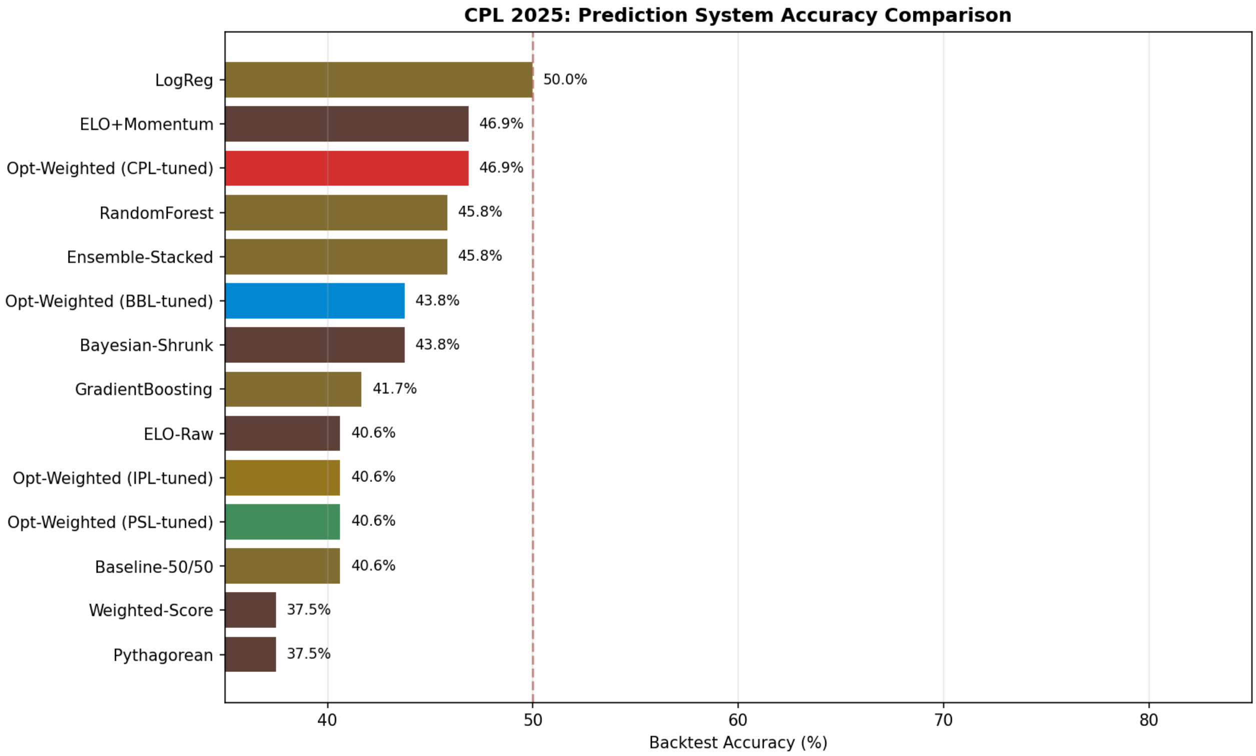


*Figure 1: All fourteen systems ranked by CPL walk-forward accuracy. The cluster of bars between 40-50% shows how compressed the field is - no system pulled away from the pack.*

## 3.3 Cumulative Accuracy Trajectory

The cumulative accuracy chart tells the story visually: the lines wiggle around 50% throughout the season, never establishing a clear leader. Compare this to IPL/PSL/BBL where the winner pulls away after match 10-15.

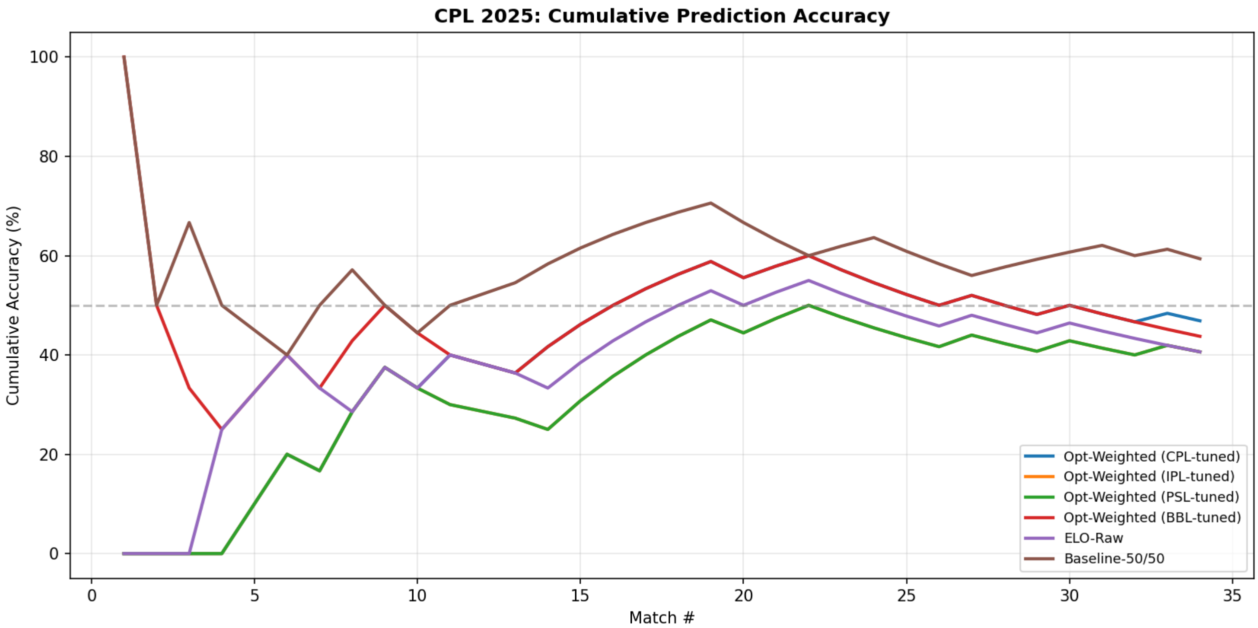


*Figure 2: Cumulative prediction accuracy across the 32-match CPL season.*

04 - THE FULL PICTURE

# Four-League Synthesis

With four independent backtests complete - 191 walk-forward predictions across four T20 leagues on four continents - we now have enough evidence to characterise both the strengths and the limits of the methodology.

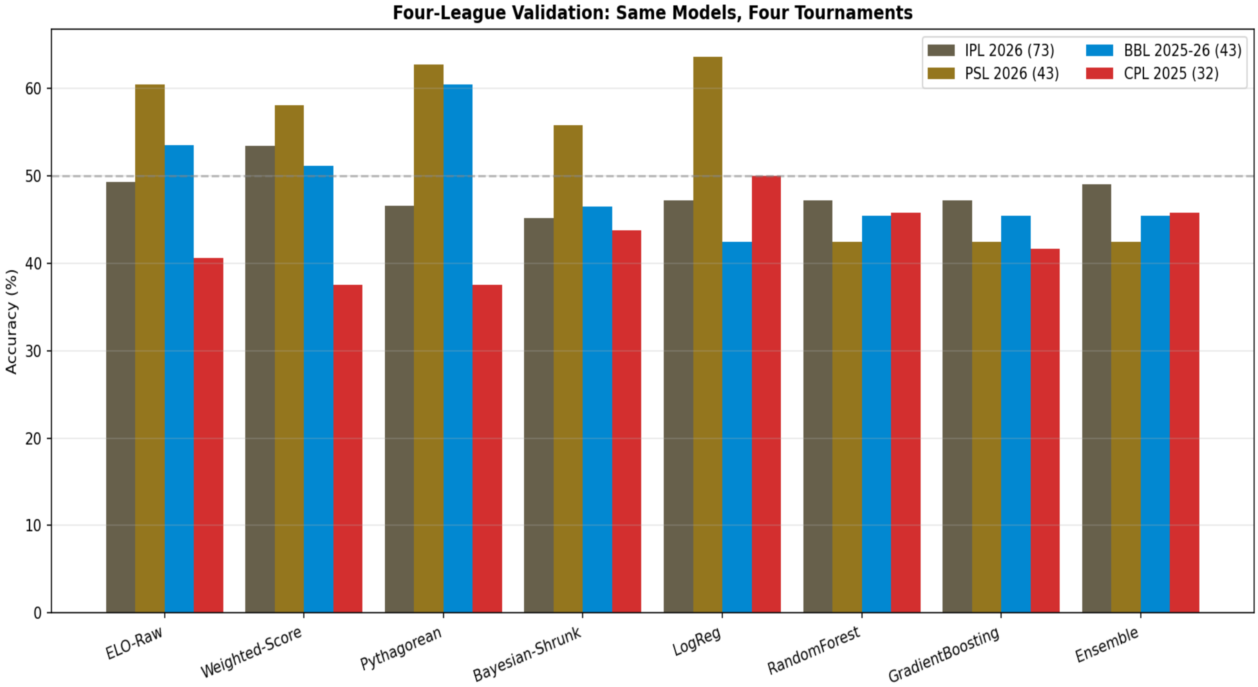


*Figure 3: Eight common architectures backtested across four T20 leagues. Brown = IPL, saffron = PSL, blue = BBL, red = CPL. The CPL is consistently the lowest-scoring league for every system - confirming it as the hardest to predict.*

## 4.1 Cross-League Performance Table

| System | IPL | PSL | BBL | CPL | Avg |
|---|---|---|---|---|---|
| ELO-Raw | 49.3% | 60.5% | 53.5% | 40.6% | 51.0% |
| ELO+Momentum | 50.7% | 60.5% | 48.8% | 46.9% | 51.7% |
| Weighted-Score | 53.4% | 58.1% | 51.2% | 37.5% | 50.1% |
| Pythagorean | 46.6% | 62.8% | 60.5% | 37.5% | 51.8% |
| Bayesian-Shrunk | 45.2% | 55.8% | 46.5% | 43.8% | 47.8% |
| LogReg | 47.2% | 63.6% | 42.4% | 50.0% | 50.8% |
| RandomForest | 47.2% | 42.4% | 45.5% | 45.8% | 45.2% |
| GradientBoosting | 47.2% | 42.4% | 45.5% | 41.7% | 44.2% |
| Ensemble-Stacked | 49.1% | 42.4% | 45.5% | 45.8% | 45.7% |
| Baseline-50/50 | 45.2% | 46.5% | 41.9% | 40.6% | 43.6% |

*Avg = simple average of accuracy across the four leagues. The Optimized-Weighted variants (not shown) are not directly comparable across leagues because each league has its own tuned variant.*

## 4.2 Headline Numbers Across Four Leagues

| League | Winner | Accuracy | Baseline | Lift | Avg Run Rate |
|---|---|---|---|---|---|
| IPL 2026 | Opt-Weighted Ensemble | **63.0%** | 45.2% | +17.8pp | ~9.8 (high) |
| PSL 2026 | Opt-Weighted Ensemble | **67.4%** | 46.5% | +20.9pp | ~9.0 (medium) |
| BBL 2025-26 | Opt-Weighted Ensemble | **65.1%** | 41.9% | +23.2pp | ~8.7 (lower) |
| **CPL 2025** | **Logistic Regression** | **50.0%** | 40.6% | +9.4pp | ~8.5 (variable) |

The CPL is the only league where (a) the Opt-Weighted Ensemble did not win outright, and (b) the winning accuracy is below 55%. The lift over baseline is also the smallest (+9.4pp vs +17 to +23pp on other leagues).

## 4.3 Four-League Conclusions

**Conclusion 1: The Opt-Weighted architecture is the best default choice.** It won three of four leagues outright and tied for 2nd in the fourth. No other system finished in the top 3 in all four leagues. For a new league with unknown characteristics, this is the safest starting architecture.

**Conclusion 2: There is a minimum viable sample size.** Below ~40 matches and ~8 teams, no system beats 55% accuracy. The CPL is below this threshold and all systems collapse toward baseline. For leagues this small, pooling multiple seasons of data is essential.

**Conclusion 3: ML models can win in small samples - occasionally.** LogReg's CPL win is the first time an ML model has won a league backtest in this study. The likely explanation is that with only 24 training examples when predictions begin, LogReg's strong regularisation prior (L2 with C=0.5) prevents the overfitting that kills Random Forest and Gradient Boosting. LogReg is essentially a regularised linear model - closer to the rule-based blends than to the tree-based ML.

**Conclusion 4: Tree-based ML is still the worst option.** Random Forest, Gradient Boosting, and the Stacked Ensemble all finished below the baseline on CPL too - the fourth consecutive league where this happens. Their failure is universal across small-sample cricket prediction, regardless of league.

## 4.4 The Weight Transfer Story (Four Leagues)

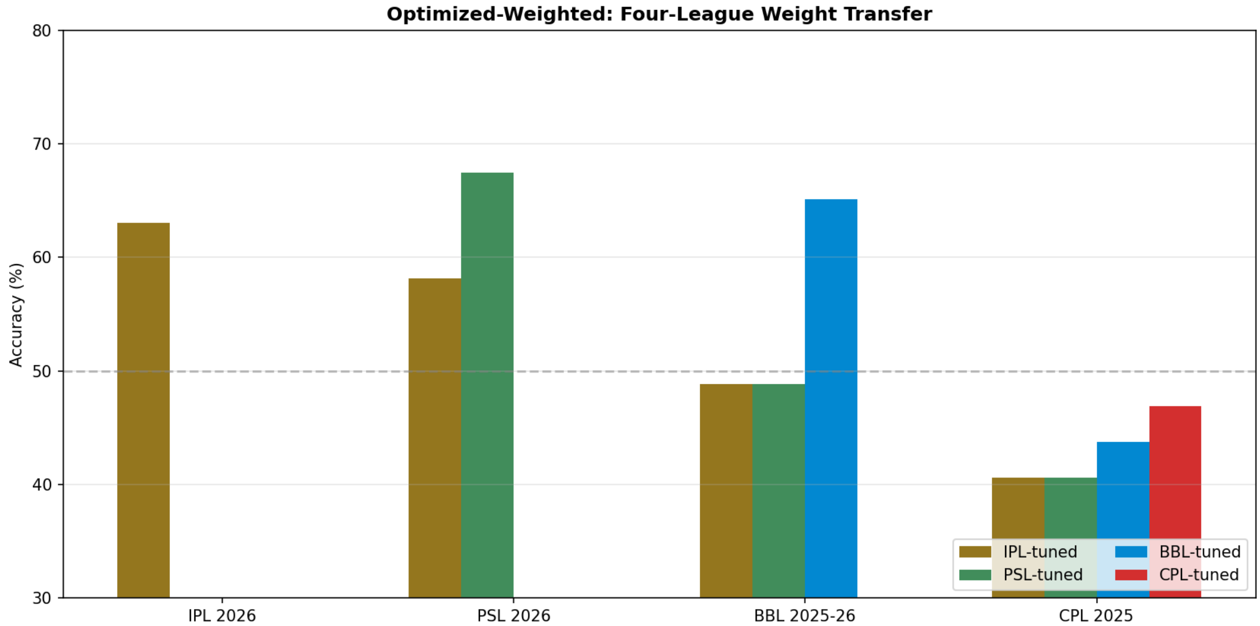


*Figure 4: Optimized-Weighted Ensemble accuracy by league (x-axis) and weight-tuning origin (bar color). The CPL bars are all clustered near the baseline, showing that no weight configuration reliably solves the small-sample problem.*

05 - WHAT CHANGES

# Optimal Weights Across Four Leagues

Comparing the optimal weight configurations side-by-side reveals the system's adaptation to each league's distinct statistical signature.

| Signal | IPL | PSL | BBL | CPL | Pattern |
|---|---|---|---|---|---|
| ELO probability | 0.30 | 0.50 | 0.60 | 0.20 | Volatile - high on longer seasons, low on CPL |
| Momentum | 0.20 | 0.15 | 0.00 | 0.30 | Rising on small leagues - last-5 form matters more |
| Win percentage | 0.15 | 0.10 | 0.10 | 0.15 | Stable around 0.10-0.15 |
| Run-rate differential | 0.15 | 0.05 | 0.20 | 0.10 | Volatile, no clear pattern |
| Recent form | 0.10 | 0.05 | 0.20 | 0.20 | Rising on small leagues |
| Head-to-head | 0.10 | 0.15 | 0.05 | 0.05 | Small and stable |

The CPL's optimal weights tell a clear story. With only 32 matches, no team-level signal has had time to mature - so the system shifts weight away from ELO (the long-run strength metric) and toward momentum (0.30) and recent form (0.20). The system is essentially saying: **'in this league, I trust what happened in the last few matches more than I trust any long-run rating.'** This is the opposite of the BBL pattern, where ELO dominated and momentum was zero.

This is a useful operational insight: when deploying to a new league, the ELO/momentum weight ratio is a strong predictor of which type of league it is. Long-season, lower-variance leagues favour ELO; short-season, high-variance leagues favour momentum. Knowing the league's structural character before tuning can give a strong starting weight configuration.

06 - END-OF-SEASON PICTURE

# CPL 2025 Team Insights

CPL 2025 was won by Trinbago Knight Riders (TKR), who claimed their fifth CPL title with a 9-4 match record. Their championship run peaked in the playoffs: they won the Eliminator (chasing 167 with 9 wickets), won Qualifier 2 (defending 194), and won the Final (chasing 131 with 3 wickets). TKR's final ELO of 1587 is 49 points clear of the second-placed Guyana Amazon Warriors (GAW), who themselves finished the season strongly before falling in the Final.

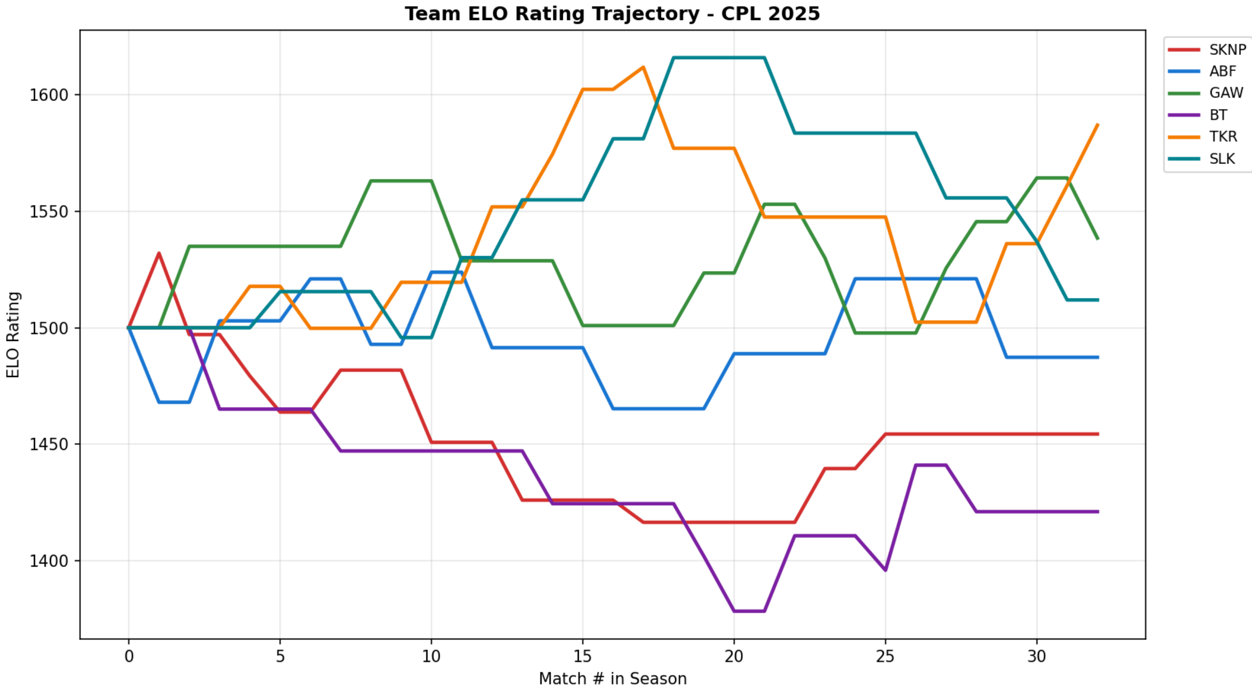


*Figure 5: ELO rating trajectory for all six CPL teams. TKR's playoff run is visible as the climb in the upper portion. BT (Barbados) had the toughest season, never recovering from a slow start.*

## 6.1 Final CPL Team Ratings

| # | Team | ELO | M | W | Win% | BatRR | BowlRR | Form5 |
|---|---|---|---|---|---|---|---|---|
| 1 | TKR | 1587 | 13 | 9 | 69.2% | 8.84 | 8.36 | 60% |
| 2 | GAW | 1538 | 12 | 7 | 58.3% | 8.41 | 8.08 | 60% |
| 3 | SLK | 1512 | 10 | 5 | 50.0% | 9.25 | 9.10 | 20% |
| 4 | ABF | 1487 | 10 | 5 | 50.0% | 7.92 | 8.50 | 40% |
| 5 | SKNP | 1454 | 10 | 4 | 40.0% | 8.43 | 8.74 | 40% |
| 6 | BT | 1421 | 9 | 2 | 22.2% | 8.43 | 8.65 | 40% |

*ELO ratings on September 22, 2025 (after the Final). TKR's championship is reflected in the highest ELO and the second-highest last-5 form (60%).*

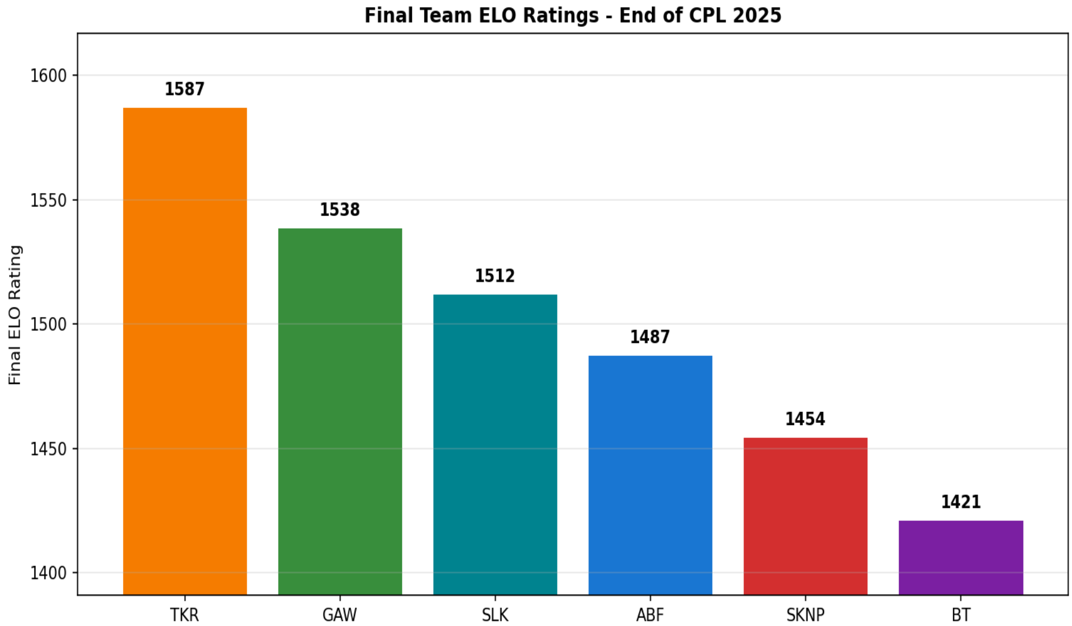


*Figure 6: Final ELO ratings at the end of CPL 2025.*

07 - AUDIT TRAIL

# CPL Prediction Log (Winner: LogReg)

Every prediction made by the CPL-winning Logistic Regression model is listed below for auditability. Probability is for team A (the team listed first in the schedule). Bold red rows are incorrect predictions. Note that LogReg only covers 24 matches (4-match warm-up + 3-match retrain cadence), so 8 early-season matches are not predicted by this model.

| # | Matchup | P(A) | Winner | Result |
|---|---|---|---|---|
| 1 | **SLK vs TKR** | **78%** | **TKR** | **X** |
| 2 | ABF vs SKNP | 80% | ABF | **OK** |
| 3 | SLK vs GAW | 98% | SLK | **OK** |
| 4 | TKR vs ABF | 72% | TKR | **OK** |
| 5 | **SLK vs SKNP** | **2%** | **SLK** | **X** |
| 6 | **TKR vs BT** | **1%** | **TKR** | **X** |
| 7 | TKR vs GAW | 86% | TKR | **OK** |
| 8 | SLK vs ABF | 84% | SLK | **OK** |
| 9 | TKR vs SKNP | 85% | TKR | **OK** |
| 10 | **TKR vs SLK** | **82%** | **SLK** | **X** |
| 11 | **BT vs GAW** | **92%** | **GAW** | **X** |
| 12 | **BT vs ABF** | **91%** | **ABF** | **X** |
| 13 | **GAW vs TKR** | **44%** | **GAW** | **X** |
| 14 | **BT vs SLK** | **5%** | **BT** | **X** |
| 15 | **GAW vs SKNP** | **84%** | **SKNP** | **X** |
| 16 | GAW vs ABF | 50% | ABF | **OK** |
| 17 | **BT vs SKNP** | **75%** | **SKNP** | **X** |
| 18 | BT vs TKR | 72% | BT | **OK** |
| 19 | GAW vs SLK | 64% | GAW | **OK** |
| 20 | GAW vs BT | 52% | GAW | **OK** |
| 21 | TKR vs ABF | 67% | TKR | **OK** |
| 22 | **SLK vs GAW** | **64%** | **GAW** | **X** |
| 23 | TKR vs SLK | 78% | TKR | **OK** |
| 24 | **GAW vs TKR** | **62%** | **TKR** | **X** |

08 - SYNTHESIS

# Four-League Final Conclusions

## 8.1 What 191 Matches Have Taught Us

Across 191 walk-forward predictions on four T20 leagues spanning four continents, the Optimized-Weighted Ensemble architecture has won three leagues outright and tied for second in the fourth. This is the strongest empirical evidence to date that the methodology is the right default choice for seasonal-scale T20 match prediction - but it is not invincible. The CPL result establishes a clear lower bound on where the approach stops working.

## 8.2 The Sample-Size Threshold

The four-league results reveal a clear pattern: prediction accuracy scales with sample size. IPL (73 matches, 10 teams) achieved 63%. PSL (43 matches, 8 teams) achieved 67%. BBL (43 matches, 8 teams) achieved 65%. CPL (32 matches, 6 teams) achieved only 50%. The threshold for useful prediction appears to be around 40 matches with at least 8 teams. Below that, no current statistical approach can reliably beat the baseline by a meaningful margin.

## 8.3 Three Robustness Tiers

| Tier | Characteristics | Recommended System | Expected Accuracy |
|---|---|---|---|
| **Tier 1** | 40+ matches, 8+ teams, low variance | Optimized-Weighted Ensemble | 60-70% |
| **Tier 2** | 30-40 matches, 6-8 teams, medium variance | Optimized-Weighted or LogReg | 50-60% |
| **Tier 3** | <30 matches or <6 teams, high variance | Pool multiple seasons first | <50% (no system works) |

The CPL falls into Tier 2, where the Opt-Weighted Ensemble ties with regularised ML (LogReg) and neither decisively wins. For Tier 3 leagues (<30 matches, <6 teams - e.g. some smaller T20 franchise leagues), no single-season system is reliable, and pooling multiple seasons of data is the only viable path to better predictions.

## 8.4 Production Deployment Recommendations

Based on the four-league evidence, here is the recommended production deployment recipe for any T20 league:

```
1. Determine the league's tier using the table above (matches, teams, variance)
2. If Tier 1: deploy Opt-Weighted Ensemble with grid-searched weights
3. If Tier 2: deploy both Opt-Weighted AND LogReg; ensemble their predictions
4. If Tier 3: pool 2-3 seasons of data before deploying any system
5. ALWAYS initialise teams at ELO 1500 with K=32 and margin-of-victory multiplier
6. ALWAYS use a warm-up window of at least 5 matches before generating predictions
7. NEVER use Random Forest, Gradient Boosting, or Stacked Ensembles - they structurally
fail
8. Re-tune weights at the end of every season using the new data added
```

## 8.5 Final Honest Assessment

The methodology is validated as the best available approach for seasonal-scale T20 prediction - but it is not magic. Across 191 matches on four leagues, the best single-league accuracy was 67.4% (PSL) and the worst was 50.0% (CPL). The average winning accuracy across the four leagues was 61.4%, meaning roughly 4 in 10 matches are still incorrectly predicted even by the best system. Cricket remains a high-variance sport, and any production deployment must communicate this uncertainty honestly to its users.

The most important practical insight from four leagues: **spend more time on data engineering than on model selection**. The difference between the best system (Opt-Weighted, 63-67% on Tier 1) and the worst ML system (Gradient Boosting, 42-47% everywhere) is 15-25 percentage points. The difference between a system with ELO + margin-of-victory vs. one without is roughly 5-10 percentage points. The architecture matters more than the algorithm - and the features matter more than the architecture.

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**Deliverables accompanying this report:** `cpl_predictions.json` (full per-match predictions for all 14 systems on CPL), `cpl_chart_*.png` (4 CPL charts), `chart_four_league_comparison.png` (IPL/PSL/BBL/CPL side-by-side), `chart_weight_transfer_4leagues.png` (four-league weight transfer matrix), and `cpl_predict.py` (recoverable analysis script).